The Monthly Physics Quest

Quest #001 · November 2026

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NGC 1385 | Chandra · Hubble · Webb.

Logistics & Guidelines

Date Issued:	November 1, 2026
Submission Deadline:	November 30, 2026 (23:59 UTC)
Submission Format:	A single typed or legibly scanned PDF via email.

- Solutions must be prepared in English and transmitted via email to m.shokouhi2001@gmail.com exclusively as a PDF attachment. The email subject line must strictly adhere to the following convention:

  [MPQ] Solution: Quest 001 – [First Name Last Name]

  Submissions delivered via alternative channels or in non-compliant formats will not be evaluated.
- The header or title block of the solution document must clearly indicate the participant's full name, institutional affiliation (school/university), and date of submission.
- Solutions should not introduce unsubstantiated assumptions beyond the problem statement. If supplementary hypotheses or physical idealizations are necessary, they must be explicitly stated and justified. All theoretical results must be derived systematically from first principles, and any approximations made must be clearly identified with an estimation of validity.
- While conceptual and theoretical discussions among participants are permitted, all analytical derivations, computations, and final manuscripts must represent independent, original work.
- Participants may consult external literature (e.g., monographs, textbooks, and peer-reviewed journals). All referenced works must be properly credited using a consistent academic citation standard.
- The use of artificial intelligence (AI) tools is permitted solely for literature retrieval, symbolic/numerical computing, code prototyping, and data visualization. Any AI assistance must be transparently declared in the solution, explicitly specifying the tool and model version along with its exact role in the derivation. Furthermore, an unedited log of the prompt–response transcript must be submitted as a supplementary PDF document.
- Official solution and final scores will be published following the conclusion of the review period.

Triaxial Conducting Ellipsoid in a Uniform Electric Field

Consider a solid insulating ellipsoid of uniform volume charge density ρ, centered at the origin, with semi-principal axes a_1, a_2, and a_3 aligned along the Cartesian coordinate axes x_1, x_2, and x_3, respectively.

1. Determine the electric field $\boldsymbol{E}(\boldsymbol{x})$ at an arbitrary point $\boldsymbol{x}$ *inside* the ellipsoid.

Now, consider an isolated, uncharged conducting ellipsoid possessing the identical geometry, immersed in a uniform external electric field $\boldsymbol{E}_0$.

2. Determine the induced surface charge density $\sigma(\boldsymbol{x})$ on the conducting ellipsoid, and compute its total induced electric dipole moment $\boldsymbol{p}$.

3. Using the general result derived in Part 2, obtain explicit, closed-form expressions for the surface charge density in the following limiting geometries:

$$
\begin{aligned}
&a_1 < \infty, \quad a_2 < \infty, \quad a_3 \to \infty, && \boldsymbol{E}_0 = E_0 \cos(\alpha)\hat{\boldsymbol{x}}_1 + E_0 \sin(\alpha)\hat{\boldsymbol{x}}_2; \\
&a_1 = a, \quad a_2 = a, \quad a_3 = b > a, && \boldsymbol{E}_0 = E_0\hat{\boldsymbol{x}}_1; \\
&a_1 = a, \quad a_2 = a, \quad a_3 = b < a, && \boldsymbol{E}_0 = E_0\hat{\boldsymbol{x}}_1.
\end{aligned}
$$

4. Provide quantitative graphical representations of the surface charge density for the following specific aspect ratios:

$$
\begin{aligned}
&a_1 = a, \quad a_2 = 2a, \quad a_3 = 3a, && \boldsymbol{E}_0 = E_0\hat{\boldsymbol{x}}_1; \\
&a_1 = a, \quad a_2 = a, \quad a_3 = 2a, && \boldsymbol{E}_0 = E_0\hat{\boldsymbol{x}}_1; \\
&a_1 = 2a, \quad a_2 = 2a, \quad a_3 = a, && \boldsymbol{E}_0 = E_0\hat{\boldsymbol{x}}_1.
\end{aligned}
$$